ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO TECHNICAL PROGRAMMING

PRACTICE EXAM 6 (PAPER 2) · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Introduction to Technical Programming practice exam 6 (paper 2).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Programs are not marked strictly on syntax: give marks for correct logic, correct variables and correct output. Accept other variable names, other prompts and other correct ways of solving the problem.

QUESTION 1

1.1 Project created and named correctly ✓ (1)

1.2 Suitable sprite ✓ (1)

1.3 Suitable backdrop ✓ (1)

1.4 Five variables created ✓✓✓✓✓ (distance, passengers, fare, total, trip / or similar) and set to 0 at the start ✓✓✓ (8)

1.5 repeat 3 ✓ block around the trip blocks ✓ (2)

1.6 ask block ✓ with a clear question ✓; set distance to answer ✓ (3)

1.7 ask block ✓ with a clear question ✓; set passengers to answer ✓ (3)

1.8 set fare to ✓ 12 + ✓ (8 × distance) ✓ (3)

1.9 if ✓ passengers > 4 ✓ then ✓ change fare by 10 ✓ (4)

1.10 say ✓ the fare (joined with text) ✓ (2)

1.11 change total by fare ✓ (1)

1.12 say total after the loop ✓ (1)

1.13 Fares R52, R102 and R36 ✓; total R190 ✓ (2)

```
when green flag clicked
set [distance] to 0
set [passengers] to 0
set [fare] to 0
set [total] to 0
set [trip] to 0
repeat 3
    change [trip] by 1
    ask [How many kilometres?] and wait
    set [distance] to (answer)
    ask [How many passengers?] and wait
    set [passengers] to (answer)
    set [fare] to (12 + (8 * distance))
    if < (passengers) > 4 > then
        change [fare] by 10
    say (join [Fare: R] (fare)) for 2 seconds
    change [total] by (fare)
end repeat
say (join [Total of all fares: R] (total)) for 3 seconds
```

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QUESTION 2

2.1 File created and named correctly ✓ (1)

2.2 #include <iostream> ✓ (and <string>); using namespace std; ✓ (2)

2.3 int main() ✓; rpm, diameter, speed declared as double ✓✓, category as string ✓ (4)

2.4 Two prompts with cout ✓✓; two cin statements into the correct variables ✓✓ (4)

2.5 speed = ✓ 3.14159 * diameter * rpm ✓ / 60 ✓ (3)

2.6 if (speed < 0.5) ✓ category = "Slow" ✓; else if (speed < 1.5) ✓✓ category = "Normal" ✓; else ✓ category = "Fast" ✓; correct structure ✓ (8)

2.7 Display the speed ✓ with the unit ✓ and the category ✓ (3)

```
// YourName - Question 2
#include <iostream>
#include <string>
using namespace std;

int main() {
    double rpm = 0, diameter = 0, speed = 0;
    string category = "";

    cout << "Enter the motor speed (rpm): ";
    cin >> rpm;
    cout << "Enter the wheel diameter (m): ";
    cin >> diameter;

    speed = 3.14159 * diameter * rpm / 60;

    if (speed < 0.5) {
        category = "Slow";
    } else if (speed < 1.5) {
        category = "Normal";
    } else {
        category = "Fast";
    }

    cout << "Speed: " << speed << " m/s" << endl;
    cout << "Category: " << category << endl;
    return 0;
}
```

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QUESTION 3

3.1 File created and named correctly ✓ (1)

3.2 #include <iostream> ✓; using namespace std; ✓ (2)

3.3 int main() ✓; days ✓, total and average (double) ✓, max / wettest day / dry counter ✓ declared (4)

3.4 Prompt ✓ and cin ✓ inside a loop ✓ that repeats while days < 1 or days > 31 ✓ (4)

3.5 Array of type double ✓ with size 31 ✓ named correctly ✓; declared before use ✓ (4)

3.6 for loop ✓ from 0 to days − 1 ✓; prompt with the day number (i + 1) ✓✓; cin into rain[i] ✓; correct use of the index ✓ (6)

3.7 total set to 0 ✓; loop ✓ adding ✓ each rain[i] ✓ (4)

3.8 max starts at rain[0] (or 0) ✓ with wettest = 1 ✓; loop through the array ✓; if rain[i] > max ✓ then max = rain[i] ✓✓ and wettest = i + 1 ✓; correct result ✓ (8)

3.9 Counter set to 0 ✓; if rain[i] == 0 ✓ the counter increases ✓ inside the loop ✓ (4)

3.10 average = total / days ✓✓ (2)

3.11 Total ✓, average ✓, wettest day with its rainfall ✓ and dry days ✓ displayed (4)

```
// YourName - Question 3
#include <iostream>
using namespace std;

int main() {
    int days = 0, dryDays = 0, wettestDay = 1;
    double rain[31];
    double total = 0, average = 0, maxRain = 0;

    do {
        cout << "Enter the number of days (1-31): ";
        cin >> days;
    } while (days < 1 || days > 31);

    for (int i = 0; i < days; i++) {
        cout << "Rainfall for day " << i + 1 << " (mm): ";
        cin >> rain[i];
    }

    maxRain = rain[0];
    for (int i = 0; i < days; i++) {
        total = total + rain[i];
        if (rain[i] > maxRain) {
            maxRain = rain[i];
            wettestDay = i + 1;
        }
        if (rain[i] == 0) {
            dryDays++;
        }
    }
    average = total / days;

    cout << "Total rainfall: " << total << " mm" << endl;
    cout << "Average rainfall: " << average << " mm per day" << endl;
    cout << "Wettest day: day " << wettestDay << " with " << maxRain << " mm" << endl;
    cout << "Dry days: " << dryDays << endl;
    return 0;
}
```

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TOTAL: 100

END OF MARKING GUIDELINE